

MUHAMMAD ABDULLAH

Faisalabad, Pakistan

+923270030506 | muhammad.abdullah.datatech@gmail.com

[LinkedIn](#) | [GitHub](#)

Professional Summary

Data Science graduate with hands-on experience building, validating, and deploying end-to-end machine learning, clustering, and data-driven systems. Proficient in Python, R, SQL, statistical hypothesis testing, unsupervised learning, and predictive modelling. Demonstrated ability to translate complex data into operational impact by developing enterprise ERP solutions, deploying live production ML applications, and executing advanced data cluster management assignments. Certified IT Specialist in Data Analytics (Certiport/Pearson VUE). Seeking an entry-level Data Scientist role to drive business growth and operational efficiency through rigorous modelling and insights.

Projects**Supply Chain Integrated ERP Project | *Infinitivs* (2026)**

- **Designed and integrated an end-to-end ERP data pipeline** to streamline supply chain workflows, inventory management, and procurement tracking.
- **Optimized data querying and reporting efficiency** using advanced SQL and Python backend logic to centralize disparate operational metrics into unified dashboards.
- **Automated inventory forecasting and stock monitoring workflows**, reducing manual reconciliation time and improving real-time visibility for decision-makers.

Customer Churn Prediction — End-to-End ML Application | *Academic Project* (2026)

- **Engineered 20+ features** from a 7,000-record telecom dataset through a robust data cleaning and exploratory data analysis pipeline, uncovering the top 8 attrition drivers.
- **Outperformed baseline models by 12+ F1 points** using an optimized Random Forest classifier, validated rigorously through cross-validation and statistical hypothesis testing on feature significance.
- **Deployed a live production web application** using Streamlit with sub-second response times, delivering actionable, production-ready churn predictions and documentation to non-technical stakeholders.
- **Built an interactive monitoring dashboard** tracking real-time churn-risk indicators and dynamic feature importances to simulate enterprise marketplace health monitoring workflows.

Customer Segmentation via K-Means Clustering | *Academic Project (R)* (2024)

- **Applied unsupervised learning algorithms** in R (cluster, factoextra, ggplot2) to segment retail customer databases based on purchasing habits and demographic profiles.
- **Determined optimal cluster counts** by combining the Elbow Method and Silhouette Analysis, maximizing intra-cluster similarity and inter-cluster separation.
- **Generated actionable customer personas** to guide targeted marketing campaigns and promotional strategies based on distinct behavioral clusters.

Hierarchical Clustering & Anomaly Detection | *Academic Project (R)* (2025)

- **Implemented agglomerative hierarchical clustering** on multi-variable health and socioeconomic datasets to categorize complex behavioral groups.
- **Performed Principal Component Analysis (PCA)** for dimensionality reduction, improving cluster visualization and mitigating multicollinearity issues.
- **Identified structural anomalies and outliers** within high-dimensional datasets using distance metrics and dendrogram cutting techniques.

Training & Simulations

Data Science Job Simulation | *Forage — British Airways (2024)*

- **Built an end-to-end classification pipeline** using Python, pandas, scikit-learn, and SQL-style data manipulation on real British Airways operational datasets to improve customer booking prediction accuracy.
- **Translated model outputs into strategic business recommendations** for simulated product and operations teams using hypothesis testing, probability analysis, and structured reporting.
- **Designed data-driven eligibility criteria** for premium airport lounge access by applying feature-level statistical significance testing on behavioral and transactional data.

Technical Skills

- **Languages & Querying:** Python (OOP, Functions, Scikit-learn pipelines), R (Cluster Analysis, Dplyr, Ggplot2), SQL (Joins, Aggregations, Subqueries)
- **Machine Learning & Modeling:** Classification, Regression, Unsupervised Learning (K-Means, Hierarchical Clustering), Random Forest, Feature Engineering, Model Evaluation
- **Statistics & Experimentation:** Hypothesis Testing, Probability, A/B Experiment Design, Statistical Significance, Cross-Validation, PCA
- **Data Manipulation & Visualization:** Pandas, NumPy, Matplotlib, Power BI, Advanced Excel (Pivot Tables, Functions), Data Storytelling
- **Tools & Deployment:** Streamlit, Git/GitHub, Jupyter Notebook | **Familiar:** AWS/GCP, Airflow, Spark

Education & Certifications

BSc Computer Science | Government College University Faisalabad (2022-2026)

- **GPA:** 3.19 / 4.0
- **Certification:** IT Specialist – Data Analytics | Certiport, a Pearson VUE Business